The Impact of Violence and Insecurity on Household Welfare: The Case of Afghanistan 2011-2017

A Thesis

Submitted to the Department of Economics

American University of Central Asia

By

Habib Gul Ahmadzai

In Partial Fulfillment of the Requirements

for the Degree of Bachelor of Arts in Economics

April 2024

Bishkek, Kyrgyzstan

Supervisor: Zarylbek Kudabaev, PhD

Abstract

This thesis examines the impact of violence on Afghan households' welfare, revealing it has significant influence on household welfare. Using over 60,000 observations from three waves of household level data from the Afghanistan Living Condition Survey (ALCS), I measure household welfare using two proxy indicators, the household income and expenditures. Drawing from the three waves of the Afghanistan Living Condition Survey conducted in 2011/12, 2013/14, and 2016/17 by the Central Statistics Organization (CSO), this study reveals that households who experienced significant conflict realize about 7.3% less income. Moreover, the findings show that household consumption expenditures are lower in areas that are affected by conflict. The analysis carried out in this study emphasizes the complex interactions between violence, household demographics, socioeconomic variables, and regional differences. I find that violence and household welfare highly vary from region to region, but the distribution of violence and household welfare largely overlap, with highly volatile and insecure areas realizing significantly less income and spend less in consumption expenditures. Overall, the findings highlight the need for policies that aim to address the adverse effects of conflict on Afghan households, with an emphasis on providing targeted interventions and support for impacted areas and vulnerable groups to mitigate economic losses and effectively recover.

Acknowledgement

I would like to express my sincere gratitude to all those who have contributed to the completion of this thesis. First and foremost, I am deeply grateful to Dr. Zarylbek Kudabaev who has not only been my supervisor but has also been great professor throughout my four years of studies, and I thank him for their invaluable guidance, support, and encouragement throughout. Their expertise and constructive feedback have been instrumental in shaping this study. I acknowledge the financial support provided by the U.S. Embassy in Kabul and Afghan Student Scholarship Program for granting me scholarship, which made this research possible.

Additionally, I am grateful to my family and friends for their unwavering support and understanding during this journey, particularly my brother, Dr. Hayatullah Ahmadzai. Being far from my parents, siblings, and home country for over four years has been challenging, but I consider you my closest friends, who have greatly contributed to my experience. Despite the distance, your support has helped me overcome numerous challenges, and for that, I am forever thankful.

Table of Contents

Abstract i

Acknowledgement ii

Introduction 1

Literature Review 3

Data 9

Econometrics Model and Methodology 14

Empirical Results and Findings 18

Conclusion 29

References 32

Introduction

Afghanistan, a low-income country, has experienced devastating economic downturns throughout the past few decades that have had a significant effect on the nation's socio-economic situation and wellbeing. Various conflicts and downturns in the economy have devastated the country's economic sectors for more than 40 years and have also slowed the country's development and overall well-being. These setbacks are the results of decades of protected conflicts and violence, weak governance and corruption, dependency on aid on foreign aid, limited infrastructure, and low human capital development. Among these factors, conflict has had a lasting impact and a crucial role in determining Afghan household welfare. To explore this lasting impact and loss of livelihoods in the context of Afghanistan, empirical research is needed to inform and guide future policies targeting such adverse impacts.

The Majority of the previous economic literature has consistently demonstrated the negative effects of conflict and political unrest, attributing them to, among other things, the damage of infrastructure, displacement, loss of human capital, and the abandoning of agricultural fields. Justino (2011) examines the influence of various dimensions of violence, household demographics, socioeconomic factors, and geographical differences on household wellbeing. The paper delves into the complexities of the relationship between violence and household welfare, noting that in Afghanistan, the link may not be straightforward due to the common inflow of foreign aid, especially in highly afflicted areas with the presence of foreign forces. Therefore, violence could impact household well-being in both positive and negative ways. This paper explores and theorizes key transmission mechanisms from armed conflict to household welfare, along with shifts in the economic, social, and political behavior of households directly and indirectly affected by the violence stemming from armed conflict.

The historic cost of the Afghan conflict is substantially large. Barrett, (2018) explore the macroeconomic impacts of conflict and estimates government revenues lose due to conflict in Afghanistan since 2005. Their results showed a total revenue loss since 2005 of $3bn, and suggested that future revenue gains from peace of about 6 percent of GDP per year. At the micro level, studies find robust evidence that in Afghanistan levels of conflict and household food security are negatively correlated, arguing that a 1%increase in fatalities per 10,000 inhabitants, can reduce consumption by 9.2% (D'Souza and Jolliffe, 2013). Studies in the broader literature focusing on the negative impacts of armed conflict and valence in similar context find comparable negative effects of conflict on the household welfare (Justino, 2009, Bloomberg and Hess, 2011; Arias et al., 2019; Kaila & Azad, 2019).

These observations can assist decision-makers in informing policies and strategies that reduce the underlying causes of violence and improve household resilience, therefore fostering peacebuilding, social stability, and sustainable development in a country. Given this context, this thesis will document the growing literature and employ robust data and methodology to offer critical recommendations. The recommendations drawn from the findings of the analysis in this study will help guide future policies, ensuring effective strategies for improving household welfare and minimizing economic losses. By examining the links between violence and household welfare, this study adds to the broader conversation on conflict dynamics and socioeconomic well-being in Afghanistan.

Against this backdrop, the main question that guides this study is: What effects do violence and insecurity have on household welfare in the case of Afghanistan? Using primary household level data, this study intends to add to the existing literature on the causal relationship between conflict and household's welfare outcomes while also offering perspectives that are unique to the

context of Afghanistan. Therefore, the notion that violence is a key factor that impedes social and economic development, particularly the well-being of the Afghan households serves as the primary motivating factor for this study.

Literature Review

The literature review includes a number of both theoretical and empirical studies that analyze the relations between violence and household welfare both in Afghanistan and other violence affected countries. The primary objective of all summarized previous literature is to investigate the impacts of violence on different aspects of livelihood and economy, including household welfare. Although the impacts of conflicts are well documented in the broader literature, the topic has received relatively little attention in the literature focusing on the Afghan context. The lack of access to consistent data, difficulty in conducting fieldwork and surveys, and other relevant information on the relation between violence and welfare have resulted in the shortage of published scientific works on Afghanistan. However, throughout the last decade, a few research and data collection surveys have been conducted which, to a small degree, has increased the interest of scholars to carry out research on different aspect of life in Afghanistan. This has also opened the way for exploring the connection between violence and household welfare in Afghanistan.

A greater amount of this literature has demonstrated a negative relationship between violence and welfare, emphasizing the harm that violence does to infrastructure, population dislocation, agricultural areas, and other economic costs. For instance, using provincial-level conflict data together with household-level socioeconomic data from the Afghanistan Living Condition Survey, Floreani, López-Acevedo, and Rama (2021) studied the influence of war intensity on household consumption in Afghanistan from 2007 to 2014. Their findings indicate

that while foreign assistance flows, and army presence somewhat offset the negative effect of conflict, conflict and violence still has a negative effect on household consumption.

They combine conflict data at the provincial level with socioeconomic data at the household level from the Afghanistan Living Condition Survey. The methodology employed involves regression analysis using ordinary least squares (OLS) to evaluate the association between conflict and poverty, defined by total household consumption expenditures. The study's findings offer several important perspectives. Firstly, the presence of troops and aid flows is strongly correlated with conflict severity at the province level (Floreani et al., 2021). Second, the analysis using household data shows that conflict has a negative effect on consumption, although these impacts are somewhat offset by the flow of international assistance (Floreani et al., 2021). Interestingly, the study also shows that, despite overall economic development, poverty rates tend to be greater in regions that had less war (Floreani et al., 2021). These findings underline the significance of considering the role of aid and troops in offsetting the negative impacts of violence on household welfare as well as expanding our knowledge of the complicated link between conflict, poverty, and household consumption in Afghanistan. Therefore, the paper suggests that the identification of the complex relationship between poverty rates and the intensity of violence underscores the need for further research and suggest targeted interventions to address the complex dynamics of development in conflict-affected regions, particularly Afghanistan.

Bove and Gavrilova (2014) in their research work entitled *Income and Livelihoods in the War in Afghanistan*, seek to analyze the impact of insurgency and military deployment on the livelihoods of local communities in Afghanistan. This study examines how people respond to conflicts and the deployment of the International Security Assistance Force (ISAF) using provincial monthly wage and commodities price data from 2003 to 2009. Using econometric

methods, notably linear regression analysis, the study's findings offer interesting details about the connection between military deployment, conflict events, and livelihoods (Bove & Gavrilova, 2014). According to the analysis, uprising violence does not significantly alter commodity prices, indicating that local populations already have coping mechanisms in place. The analysis also shows that military deployment is linked to rising wages and commodity prices. Curiously, it is discovered that prices are more sensitive to deployment than to direct attacks. Beyond the destruction and loss of life and altering other economic variables such as price, the outcomes of war lead to insurgency and political instability that in turn affect economic development and individual well-being (Bove & Gavrilova, 2014). Although mechanisms are available or introduced in society to reduce such effects, it is not clear what exactly such violence means for livelihood opportunities (Bove & Gavrilova, 2014). This study fills this gap by looking into how income opportunities at community level were affected by insurgency related violence as well as ISAF deployment in Afghanistan. Moreover, these findings shed light on the dynamic interaction between conflict and economic determinants and emphasize the significance of comprehending the effects of conflict occurrences and military presence on livelihoods in Afghanistan.

Additionally, to gain better understandings the impact of violence, D'Souza and Jolliffe (2013) discuss the connection between violence and food insecurity in Afghanistan, undertaking the study titled *Conflict, Food Price Shocks, and Food Insecurity: The Experience of Afghan Households*. The researchers make use of both private geo-coded data on violent occurrences and nationally representative household survey data. Bivariate and multivariate regression models as well as geographical mappings are part of the methodology used in the paper. The study's findings offer a number of significant insights. Firstly, it is discovered that armed conflict significantly contributes to the rise in food prices and an increase in food insecurity(D'Souza & Jolliffe, 2013).

Using geographical mapping, the analysis also reveals regional variations in the severity of food insecurity and conflict. Contradictory to the common belief, the study reveals that high-conflict regions do not usually have the greatest levels of food insecurity (D'Souza & Jolliffe, 2013). This implies that other elements, such as food accessibility and coping techniques, can influence the implication of violence on food security. Additionally, households in high-conflict regions witness a slight decrease in food security, perhaps mostly as a result of rising costs for basic goods. Furthermore, the findings spotlight the need for comprehensive strategies to deal with not only the immediate consequences of violence, but the underlying socio-economic vulnerabilities that contribute to food insecurity (D'Souza & Jolliffe, 2013). These findings also highlight the need for specific solutions to address the particular issues experienced by different households in different regions, and provide light on the complex connection between war, food prices, and food insecurity in Afghanistan.

Another study by Barrett (2018) entitled *The Fiscal Cost of Conflict: Evidence from Afghanistan 2005-2016*, investigates the effect of violence on government income, offering significant insights into the economic effects of conflict in Afghanistan. This study analyzes the financial damage brought on by the war throughout the investigation period by using data on provincial conflict fatalities, and to accurately capture the link between per capita mortality and domestic revenues, a dynamic panel model estimation approach is employed (Barrett, 2018). To increase the reliability of the results, three complementing techniques are used: generalized synthetic controls, instrumental variables, and ordinary least squares. The analysis estimates that the Afghan war has cost the domestic government between 10% and 13% less revenues than it would have otherwise (Barrett, 2018). Moreover, the study suggests that the fiscal peace dividend is likely to be substantial—around 50 percent of current revenues. It also emphasizes the

importance of financing peacekeeping and conflict resolution exercises to avoid such economic costs (Barrett, 2018). These findings highlight the huge economic costs associated with war and stress the significance of funding peacekeeping and conflict resolution initiatives. For policymakers and other stakeholders working to address the economic effects of war and promote sustainable development, this study offers insightful information that will help us better grasp the fiscal aspects of the conflict in Afghanistan.

In addition to being a crucial subject of discussion in the setting of Afghanistan, the topic of how violence affects household welfare can also be a crucial one in a general context to discuss. For instance, in their work *The Impact of Armed Civil Conflict on Household Welfare and Policy*, Justino, (2011) explores in a detailed manner the multiple effects that conflicts have on households' welfare. The latter are carefully classified into direct and indirect impacts that explain immediate losses such as deaths or destruction of properties as well as long term issues such as economic and social instabilities. Nevertheless, the study goes beyond this and analyzes coping mechanisms including subsistence farming or redistribution of household labor at great personal expense or development costs (Justino, 2011). They also point out the importance of dysfunctional markets and broken social connections in perpetuation of poverty within affected communities. This approach brings to light not only the stark facts affecting these households but also constructs an important case for interventions aimed at specific policies (Justino, 2011). Therefore, the research insists on adequate systems able to alleviate temporary difficulties and promote long-term healing. It is a formative paper, which gives critical advice that can be used by policy makers and development organizations when designing strategies meant to tackle immediate and future needs of families involved in cases of civil wars; hence, assisting them attain larger peace building processes together with development goals generally (Justino, 2011).

The literature explored the varied impact of violence on household welfare in Afghanistan, emphasizing the complex and multidimensional relationships at play. In this literature, violence typically has a negative impact on economic stability and livelihoods, but the presence of military forces and aid can help alleviate some of these negative effects. Despite the mixed results due to different research methods and the specific situation in Afghanistan, the studies we looked at collectively underline the significant economic and social consequences of violence, such as infrastructure destruction, disruption to agriculture, and the displacement of communities. Important points from research show that conflict not only decreases household consumption and worsens poverty but also has negative effects on government income and financial stability. However, their findings only point to aid and military support as potential solutions that can help alleviate some of these negative impacts.

It is crucial to tailor such policies and interventions to fit the unique context and dynamics of conflict in context of specific regions to be effective. In order to address the ongoing challenges in Afghanistan due to prolonged conflict, it is vital to conduct more empirical research using primary data and robust methodologies. Further research needs to focus on expanding our understanding of how violence affects economic and social outcomes, as well as investigating successful approaches for reconstruction and growth. It is crucial for academics, policymakers, and global partners to work together and in collaboration, utilizing research findings like those highlighted in recent studies, to design solutions that meet the immediate and future requirements of impacted populations. These actions are not only necessary for enhancing household well-being, but also for fostering peace and economic security in Afghanistan and other similar settings worldwide.

Data

For the analysis carried out in this thesis, the data is sourced from the Afghanistan Living Condition Surveys (ALCS) which were conducted by the National Statistic and Information Authority (NSIA) in Afghanistan during three different periods from 2011 to 2017. The first edition of the ALCS that is used in this paper was conducted in 2011/2012 to evaluate the living conditions of the Afghan local communities and help guide future policies. Upon the rapid changes in the socio-economic and political of Afghanistan, the survey was conducted again during the years 2013/14, and a year later in 2016. ALCS is the most comprehensive survey household-level survey that gathers primary information from households. The survey has been carried out for several years in all provinces of Afghanistan and provides significant information about the living conditions of people in Afghanistan. It followed different techniques used by the United Nations Food and Agriculture Organization, and in the beginning, it was called the National Risk and Vulnerability Assessment (NRVA), which delivered information on food security and the level of poverty (CSO, 2019). The survey later received financial support from the United Nations' World Food Programme (WFP) and faced transformation, expanding the depth of its approach, and its name was changed to ALCS to reflect the country's reforms and transformation. In order to maintain comparability, the ALCS kept the same questionnaire structure, reported indicator materials, training, and data collecting methods. The ALCS data is retrospective at both national and provincial level collecting data in 35 strata, for 34 provinces and *Kuchi*/nomad population (National Statistic and Information Authority, 2019).

The main goal of the ALCS survey was to gather comprehensive data on numerous sectors in order to monitor and assess the nation's economic performance. Each wave of the survey comprises more than 21,000 households. I combine three waves of the ALCS survey conducted in

three rounds from 2011 to 2017. The total sample therefore adds up to over 60,000 observations. The data includes representative national and provincial statistics. The information gathered through this survey, which includes a cross-section of household data across a 12-month period, can be used to gain a thorough knowledge of household situations throughout the year. The information acquired also creates an understanding of the household welfare and other relevant socioeconomic and demographic characteristics (CSO, 2019).

Table 1 defines various variable and shows descriptive statistics for different variables such as household income, consumption expenditures, violence, and other household characteristics (note that summary statistics for each survey round is presented in the Appendix, Table A1). Descriptive statistics were computed to give a general overview of the variables utilized in the analysis in the research paper's data section. Firstly, examining the total income, it is evident that the mean household income is approximately 139,033.36 Afghanis, with a considerable standard deviation of around 127483.31, indicating substantial variability in income levels within the dataset. The range of income spans from 5,555.55 to 1,558,441.5 Afghanis, highlighting significant income disparities among households. Similarly, the estimated mean value for household expenditures were estimated at 7,013.006 Afghani with a standard deviation of 5,083.971. Moreover, household income and expenditures show slight variations from year to year, but both increase over the period of 2011 to 2017 (see **Figure 1A** in the Appendix) during which the surveys were conducted. It is worth noting that the dependent variables were inspected for potential outliers. Graphing both dependent variables revealed that outliers were present, and therefore they were trimmed by 0.25%. This means 0.25% of the bottom and top values of the dependent variables were excluded.

Table 1.

Variables and Descriptive Statistics

Variable	Mean	Std. Dev.	Min	Max
HH income (AFN)	139,033.36	127483.31	5555.56	1,558,441.5
HH Expenditures (AFN)	7,013.006	5,083.971	320	56,000
Violence (0/1, 1=yes)	.167	.373	0	1
HH Head education (years)	2.936	4.866	0	19
HH head age (years)	42.193	13.711	10	99
Household size (count)	7.681	3.364	1	43
HH head sex (0/1, 1=Male)	.991	.092	0	1
HH head employment (0/1, 1=yes)	.791	.407	0	1
Dependency ratio	1.275	.897	0	11
Opium cultivation in district (*Jeribs*)	40.614	139.07	0	1,485
Region (0/1, 1=West Central)	.082	.275	0	1
Region: 0/1, 1=Northeastern)	.147	.354	0	1
Region: 0/1, 1= Northern)	.146	.353	0	1
Region: 0/1, 1=Western)	.098	.297	0	1
Region: 0/1, 1=South Western)	.134	.341	0	1
Region: 0/1, 1=Southern)	.108	.31	0	1
Region: 0/1, 1=Eastern)	.109	.312	0	1
Region: 0/1, 1=Central)	.175	.38	0	1
Residence code (0/1, 1=urban)	.164	.371	0	1
Residence code (0/1, 1=rural)	.808	.394	0	1
Residence code (0/1, 1=*Kuchi*)	.027	.163	0	1
Wave 1 (0/1, 1=2011/12)	.339	.473	0	1
Wave 2 (0/1, 1=2013/14)	.339	.473	0	1
Wave 3 (0/1, 1=2016/17)	.322	.467	0	1
Observations	60.479			

The variable "violence" represents the prevalence of violence within the dataset. On average, about 17% of households in the pooled sample have experienced violence, with a standard deviation of approximately 0.37. This indicates that while violence is present, its occurrence varies among households. In terms of demographics, there appears to be variation in the educational attainment levels of household heads, as evidenced by the mean household head education level which is approximately 2.94 with a standard deviation of almost 4.87. Note that in Afghanistan primary, secondary, high school, university, and postgraduate studies count as 6, 9, 12, 16, and 19 years of schooling. The average age of the household head is roughly 42 years, with a significant standard deviation suggesting a broad age range in the sample. The average household size is about 7.68 persons, with a standard deviation of approximately 3.36, indicating variability in the household size. Furthermore, with an average of almost 0.99 mean, the majority of household heads are male, indicating that household are predominately headed by men. The average dependency ratio is about 1.27, meaning that for every working-age individual there are about 1.27 dependents (i.e., individuals aged 14 or lower and elderly above 64 years of age). Average opium cultivation at the district level is 40 Jeribs with significant variations among districts in the sample ranging from 0 Jeribs (opium free) to highly opium producing (1485 Jeribs) districts. Note that Jerib is a common unit of area measurement in Afghanistan, equivalent to one-fifth of a hectare.

To focus on the variable of interest, the distribution between households experiencing violence and those that do not, **Table 2** further breaks down the household self-reported binary variable of violence by the survey year. Out of a total of 60,479 observations, 50,388 (83.31%) households are reported to experience no violence, hence reflecting the large majority under relatively stable conditions. On the other hand, 10,091 (16.67%) households have had experiences of violence in the pooled sample, an aspect that may indeed amount to a serious minority facing

adverse welfare impacts. Moreover, violence varies from year to year showing a decreasing trend over the years. That is, more household experienced violence in 2011/12 compared to 2016/17 (see **Table 2** and **Table A1** in the Appendix). In 2011/12 about 21%, in 2013/14 about 16%, and in 2016/17, about 13.3% of the households reported any experience of violence and insecurity.

However, it should be noted that this household self-reported measure for violence only shows whether the household reported experiencing violence in a particular year, it therefore does not show the overall conflict activity in the district or region. This distribution is quite key to our study, given the fact that this is a way of pointing out the necessity of analyzing the effect of violence on household welfare and probably in the differences in economic, health, and social outcomes between these two groups. The presence of violence in 16.7% of the sample signifies a huge impact on these households, which until now has been quite small in interventions, and calls for detailed analysis of policies to act on and mitigate its effects.

Table 2.

Tabulation of Violence

Wave	Violence and Insecurity		
	Violence=0	Violence=1	Total
2011/12	16,236	4,258	20,494
	79.22	*20.78*	*100.00*
2013/14	17,243	3,241	20,484
	84.18	15.82	100.00
2016/17	16,909	2,592	19,501
	86.71	*13.29*	*100.00*
Pooled sample	50,388	10,091	60,479
	83.31	*16.69*	*100.00*

Note: First row has *frequencies,* and second row has *row percentages*

Table 3 summarizes household income and expenditure according to the incidence of violence and insecurity. The mean of the total income and expenditures is substantially lower for households that experienced violence or insecurity. This shows a clear pattern in the data that violence is a key factor that affects households' welfare.

Table 3.

Mean of Total Income and Expenditures by violence

	Violence and Insecurity	
Welfare outcome	Violence=0	Violence=1
Total Income (AFN)	140,922.56	129,599.86
Total Expenditures (AFN)	7,080.654	6,675.216

Econometrics Model and Methodology

Econometrics Model

The econometric method applied in this study helps us model the impact of violence on household income and consumption expenditure (proxies by which household welfare is measured) in Afghanistan. The aim to pursue this study is rooted in the fact that insecurity, or in other words violence, plays a very determinant role in shaping the overall economic stability and the livelihoods of households. The methodological approach to evaluate this study will be based on the application of Ordinary Least Square (OLS) regression analysis. OLS has been selected because of its efficiency in giving excellent linear unbiased estimators under the classical linear regression assumptions. The regression model is carefully specified in a way that captures the effects of violence on household income conditional on a number of other factors such as the household demographic and socioeconomic characteristics that might affect this relationship. For

instance, the model takes into account household size, the demographics of the head of the household (age, sex, education level), dependency ratio, the employment status of the head of the household, as well as regional variables to control for regional variations. Each of these variables is expected to provide insight into the diverse influences on household income while quantifying the effect of violence. The analytical model has the following mathematical form and specifications:

$$\textit{Household Welfare (ltotal_income)}_{it} = \beta_0 + \beta_1(\textit{violence})_{it} + \beta_2(\textit{lhh size})_{it} + \beta_3(\textit{lhh head age})_{it} + \beta_4(\textit{lhh head sex})_{it} + \beta_5(\textit{lhh head education})_{it} + \beta_6(\textit{depratio})_{it} + \beta_7(\textit{resident code})_{it} + \beta_8(\textit{district level opium cultivation})_{it} + \beta_9(\textit{hh head employment})_{it} + \beta_{10}(\textit{wave})_{it} + \beta_{11}(\textit{region})_{it} + \varepsilon_{it}$$

A similar linear regression model can be used to see the impacts of Violence on household expenditures by taking the total expenditure of a household (ltotal_expenditures) that a household have had during the year the survey was conducted as a dependent variable. The mathematical form of the model can be specified as follow:

$$\textit{Household Welfare (ltotal_expenditures)}_{it} = \beta_0 + \beta_1(\textit{violence})_{it} + \beta_2(\textit{lhh size})_{it} + \beta_3(\textit{lhh head age})_{it} + \beta_4(\textit{lhh head sex})_{it} + \beta_5(\textit{lhh head education})_{it} + \beta_6(\textit{depratio})_{it} + \beta_7(\textit{resident code})_{it} + \beta_8(\textit{district level opium cultivation})_{it} + \beta_9(\textit{hh head employment})_{it} + \beta_{10}(\textit{wave})_{it} + \beta_{11}(\textit{region})_{it} + \varepsilon_{it}$$

Description of Variables in the Model

- *ltotal_income*: is the natural logarithm of the household total income earned during the past 12 months at the time of the survey.
- *Ltotal expenditures*: shows the natural logarithm of total expenditure of the household spend in the past twelve months.
- *Violence*: Binary Variable shows whether a household has experienced a violence and insecurity shock throughout the year at the time of the survey (0/1, 1=yes)
- *hh_head_sex*: Shows household head sex ((0/1, 1=male).
- *hh_head_age*: is the household head's age by year.
- *hh_size*: Shows the number of people living together in a household.
- *Hh_head_education*: Shows how many years of education the Household head has.
- *Depratio*: The dependency ratio is a measure of the number of dependents aged zero to 14 and over the age of 65, compared with the total population aged 15 to 64.
- *Resident Code*: The code assigned to each household and shows whether a household is located in Rural or Urban neighborhood.
- *Opium_Share*: Shows the income share that comes from Opium cultivation on a district level.
- *hh_head_employment*: is the employment status of household head and shows whether the household head was employed at the time of the survey (0/1, 1=yes).
- *Region*: summary variables that indicate household's location in one of the eight regions of Afghanistan.
- *Wave*: dummy variable that captures the years in which the survey was conducted. The data set includes three waves: 2011/2012, 2013/2024, and 2016/2017.

Methodology

The dependent variable in the above model represents household welfare outcomes including Income and consumption expenditures, measured in Afghani (AFN) for each respective household *i* in the sample. The most important variable to be taken into account is the variable of interest, violence, which is a binary variable capturing whether a household has experienced violence or insecurity in the pooled sample. Other control variables in the models have been included to account for other influences on household income in this respect: demographic factors – household size, household head age, household head Employment, household head sex, and

household head education. These all form a full set of controls for economic and geographical dependencies and include, among others, the following variables: dependency ratio, residential area code, opium cultivation at the district level, survey year or wave fixed effects, and region fixed effects. The above description of variables shows each variable included in the model with a short definition in the context of our discussion.

The estimated coefficients for the variables in the model, ($\beta_1, \beta_2, \ldots, \beta_{11}$) are estimated using Ordinary Least Squares (OLS) technique. This technique of estimation was chosen due to its efficiency in providing the best Linear Unbiased Estimators under classical regression assumptions such as no multicollinearity, homoscedasticity, and independence of errors. A negative coefficient for violence (β_1) would indicate that the experience of violence negatively influences the income of households, a signal for deteriorating economic welfare. A positive coefficient could, on the other hand, be construed as an effect of unexpected compensation through some other economic activity or resilience mechanism that would be used to offset reduction in income caused by violence. Other coefficients ($\beta_2, \beta_3 \ldots, \beta_{11}$) in model capture the specific effects that are exerted by corresponding variables on household income, holding constant all other variables. For instance, the coefficient of household head education variable captures the impact that education has on household welfare and depending on the sign of the coefficient, it can be easily determined whether it has negative or positive effects on the household welfare.

Moreover, such a broad model would also enable detailed research on the effects of violence on household welfare, using the controls for a number of demographic, economic, and geographic factors such as age, gender, regions and others. This approach identifies not only the direct impact of violence but also the role of different socio-economic factors in shaping household income. Based on the model outlined above, the interaction and unique contribution of each of

these variables will give critical insights that could inform policy decisions relevant to efforts targeting the improvement of household welfare in the conflict-affected areas of Afghanistan. Model outcomes from the study provide a good basis for discussion regarding policy measures in the lessening of the deleterious effects of violence on the economic stability of the households, which is a great addition to academic literature and practical policymaking in conflict settings.

Empirical Results and Findings

The empirical findings of the linear regression analysis offer important new understandings of how violence and insecurity affect household welfare in Afghanistan. The regression model includes multiple factors to facilitate a comprehensive understanding of the determinants of household income and consumption expenditures – the proxy variables for welfare. The estimated regression coefficient for violence is (-0.073), as indicated in **Table 4**, which is statistically significant at the 1% level according to its p-value. This implies a strong negative correlation between violence and household income. The coefficient's estimated size indicates that a one-unit rise in violence, holding other variables constant, would imply a reduction in household income by 7.3 percent. This would mean that 16.69 % of the sample size that has experienced violence, as can be seen in **Table 2**, has 7.3% less income compared to those households that had not experienced violence, in our case 83.1%. This decrease may include the expense of the acts of violence which could range from medical expenditures to more intangible expenses like not having the ability to work due to injury or detention.

Table 4.

Linear Regression Using Total Income as Dependent Variable

Log. Total Income	Coef.	St.Er.	t-value	p-value	[95% Conf	Interval]	Sig
Violence (0/1, 1=yes)	-.073	.008	-9.39	0	-.088	-.058	***
HH head education	.025	.001	41.91	0	.023	.026	***
HH Head age (years)	.003	0	14.57	0	.003	.004	***
Head employment (1=yes)	.128	.007	17.29	0	.114	.143	***
HH size (count)	.076	.001	76.94	0	.074	.078	***
Head (1=Male)	.242	.033	7.28	0	.177	.307	***
Dependency ratio	-.124	.003	-38.57	0	-.13	-.118	***
Opium cultivation (*Jeribs*)	.001	.001	-0.89	.375	.00059	-.00002	
Resident code: (2=rural)	-.247	.008	-32.76	0	-.262	-.232	***
Resident code: (3=*kuchi*)	-.185	.019	-9.94	0	-.222	-.149	***
Region: Northeastern	.422	.012	35.04	0	.399	.446	***
Region: Northern	.411	.012	35.39	0	.388	.434	***
Region Western	.388	.013	29.51	0	.362	.414	***
Region: South Western	.461	.013	35.30	0	.435	.487	***
Region: Southern	.705	.013	53.62	0	.68	.731	***
Region: Eastern	.196	.014	14.11	0	.169	.223	***
Region: Central	.579	.012	48.11	0	.555	.602	***
Wave 2 (2013/14)	.061	.007	9.02	0	.048	.074	***
Wave 3 (2016/17)	.028	.007	4.28	0	.015	.041	***
Constant	10.332	.037	280.85	0	10.26	10.404	***

Mean dependent var	11.543	SD dependent var	0.771
R-squared	0.268	Number of obs	60,479
F-test	1123.020	Prob > F	0.000
Akaike crit. (AIC)	121338.671	Bayesian crit. (BIC)	121518.872

Notes: Reference categories for wave, region, and resident code are 2011/12, west central region, and urban, respectively. Robust s*tandard errors are in parentheses. *** p<.01, ** p<.05, * p<.1*

In addition, the same can be observed when total household expenditure is taken as the proxy variable for welfare. In this case, the estimated coefficient for violence is (-.021), as indicated in **Table 4**, which is significant at the 1% level. This implies that a household that has experienced violence at the time of the survey realized approximately 2.1% less consumption expenditures as compared to those that had not experienced violence or insecurity.

Table 5.

Linear Regression Using Total Expenditure as Dependent Variable

Log. Total exp.	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
Violence (0/1, 1=yes)	-.021	.006	-3.32	.001	-.033	-.008	***
HH head education	.016	0	33.68	0	.015	.017	***
HH Head age (years)	.003	0	13.58	0	.002	.003	***
Head employment (1=yes)	.088	.006	14.95	0	.076	.099	***
HH size (count)	.083	.001	103.88	0	.081	.084	***
Head (1=Male)	.299	.025	12.04	0	.251	.348	***
Dependency ratio	-.054	.003	-21.52	0	-.059	-.05	***
Opium cultivation (*Jeribs*)	0	0	-0.29	.775	0	0	
Resident code: (2=rural)	-.308	.006	-49.52	0	-.32	-.296	***
Resident code:(3=*kuchi*)	-.342	.015	-23.36	0	-.371	-.314	***
Region: Northeastern	.199	.01	19.64	0	.179	.219	***
Region: Northern	.218	.01	22.25	0	.198	.237	***
Region Western	.138	.011	12.74	0	.116	.159	***
Region: South Western	.017	.011	1.57	.116	-.004	.038	
Region: Southern	.27	.012	23.42	0	.248	.293	***
Region: Eastern	.317	.011	29.39	0	.296	.338	***
Region: Central	.303	.01	29.36	0	.283	.323	***
Wave 2 (2013/14)	.207	.005	38.09	0	.196	.217	***
Wave 3 (2016/17)	.17	.005	31.44	0	.159	.18	***
Constant	7.504	.028	265.73	0	7.449	7.559	***

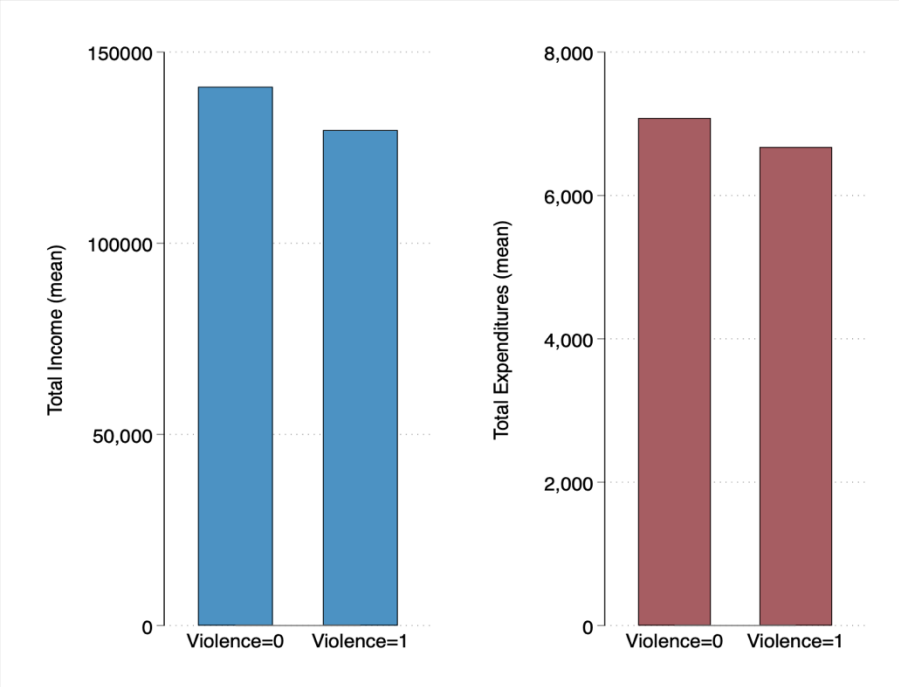
Mean dependent var	8.648	SD dependent var	0.647
R-squared	0.330	Number of obs	60479
F-test	1488.238	Prob > F	0.000
Akaike crit. (AIC)	94844.190	Bayesian crit. (BIC)	95024.391

Notes: Reference categories for wave, region, and resident code are 2011/12, west central region, and urban, respectively. Robust s*tandard errors are in parentheses. ***p<.01, **p<.05, *p<.1*

To further visualize the impact of violence estimated by the regression analysis, the following bar graphs presented in **Figure 1** have been constructed. The graphs provide a comparison of the average income and expenditures between households that experienced violence and those that did not. It can be observed that households that have experienced violence tend to have lower total income and expenditure.

Figure 1.

Bar Graph of Average Total Income and expenditures by violence

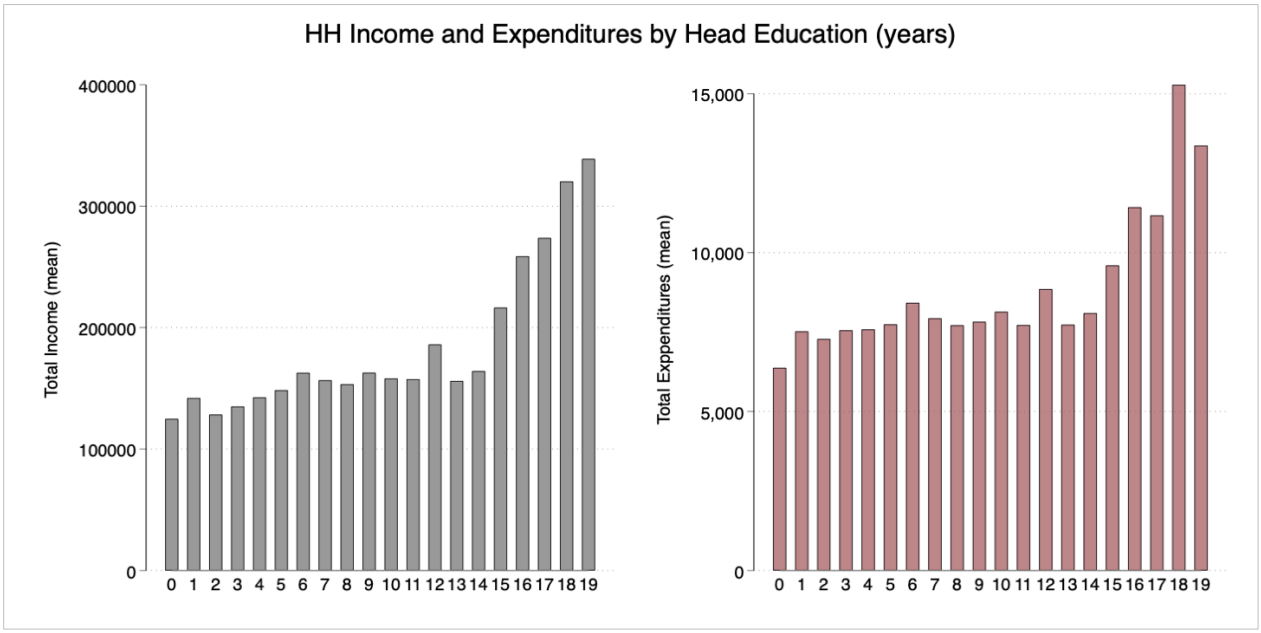


Additionally, as shown in **Table 4** and **Table 5** above, a number of factors significantly influenced household welfare (income and expenditure). For instance, another factor that emerges as a significant positive determinant of household welfare is the household head education. The estimated coefficients for the education level of the household head in both regressions for income and consumption expenditures are (0.025) and (0.16), respectively, indicating that each additional year of schooling increases household income and consumption by 2.5% and 16% respectively, holding other factors constant. **Figure 2**, further demonstrate these results, show that household income increases in educational years. The positive and statistically significant impact of education can be attributable to the human capital theory, which suggests that investing in education boosts human productivity and income which can eventually lead to the increase of a household welfare (Becker, 1962). That said, pursuing education is also linked with violence, especially in

Afghanistan where violence has significantly damaged infrastructure in educational sector and resulted in a decrease in the level of education which, in turn, is likely to have an indirect impact on the household welfare. The fact that households led by educated people often make more money emphasizes the value of education and how it affects chances for employment and other sources of income.

Figure 2.

Average Total Income by Years of Education or Schooling



Additionally, the age of the household head also shows a significant positive correlation with income and consumption expenditures, marked by a coefficient of (0.003). This means that for every additional year of the household head's age, the expected total household income and consumption expenditures increases by 3%, holding all other factors constant. This finding corroborates the notion that with each passing year, the accumulation of work experience or the mere increase in the age of the household head is associated with a substantial enhancement in earnings. Moreover, the household size has positive coefficients of (.076) and (.083) for both

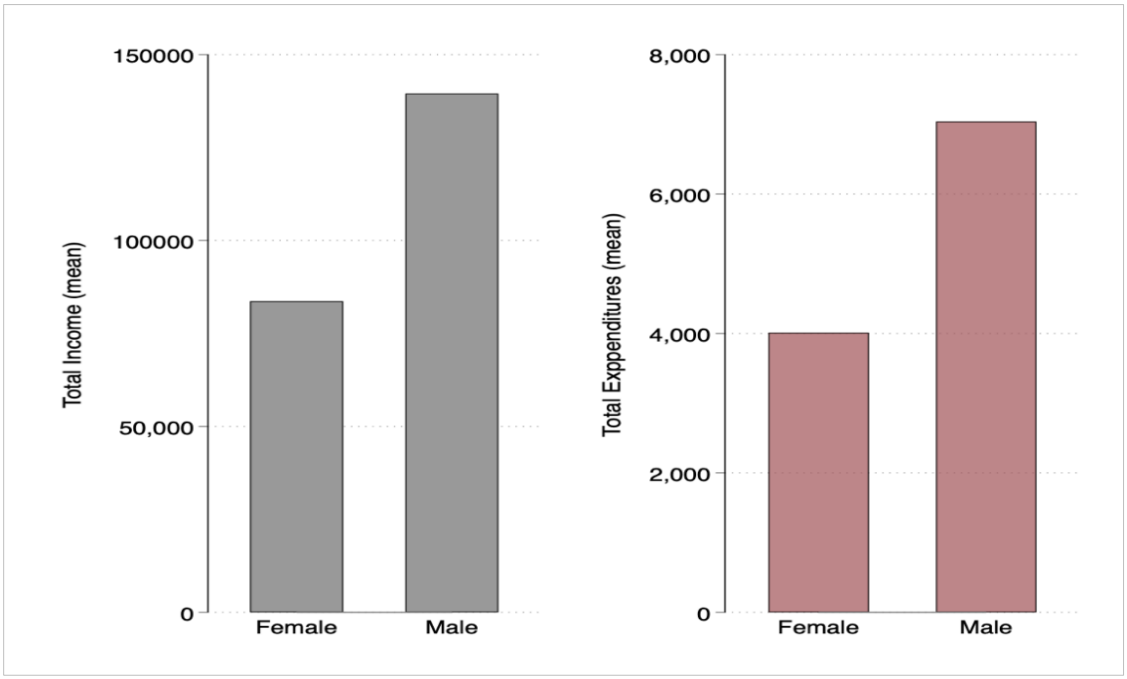
income and expenditure, respectively. This indicates a positive correlation with household welfare, suggesting that households with higher welfare are often considered to be larger. More specifically, holding other variables constant, household income is predicted to rise by 7.6% and household consumption is expected to increase by 8.3% when household size increases by one unit. This could be a result of the availability of additional labor who are actively participating in the household's income-generating activities, such as farming, which is the main economic sector in the country.

Another important variable to consider in the context of Afghanistan is the gender of the head of the household, which has a positive coefficient of (0.242)) for household income for male-headed households which implies that, holding other variables constant, households headed by males realize higher incomes than those headed by females (**Table 4**). The coefficient applies to logged total household income, implying that being a male household head is associated with a 24% higher income. Similarly, as can be seen in **Table 5,** the coefficient of household head sex is (0.299) for the household total expenditure, indicating that a household headed by male spends about 30% more in consumption. **Figure 3** below provides a visual representation of this difference in income across household by the head's gender. These coefficient produces the interpretation of cross-sectional disparities in economic outcomes in the studied in Afghanistan. More specifically, because the coefficient is both significant and positive, it indicates that there is a measurable disparity in economic welfare in male and female-headed households. Access to resources, gender-based employment discrimination, and other cultural and socio-economic variables usually accounts for such disparities in traditional Afghan societies. It is also associated to the predominant societal norms that determine what jobs and what resources of livelihood are available to men and

women. The coefficients provide a quantitative picture that helps in understanding gender inequality in relation to household welfare.

Figure 3.

Comparison of Total Household Income and Expenditures by Head of Household's Sex



Besides, the inclusion of geographical variables in the regression plays an important role in our understanding of how income is distributed geographically in Afghanistan and how it varies with different factors, for instance violence in this case, that affect income. Having this in mind, I carried out the analysis on regional level in Afghanistan and include different regions such as Eastern, Southern, South Western, Western, Northern, North Eastern, and West Central regions. Each geographical variable has a corresponding coefficient estimate that, while adjusting for other factors in the model, measures the correlation between that region and household welfare outcomes. Therefore, in comparison to the reference region (which is West-Central region), the

coefficients represent the average change in household income related to a given region's location (i.e., in the reference category).

For instance, looking at the results presented in **Table 4**, the positive coefficient (0.196) for Eastern region shows that, on average, households in the Eastern region earn about 20% more income than those in the reference region (West Central). Similarly, the positive coefficient of (0.017) for the South Western region in **Table 5** demonstrates that South Western region has 1.7% higher consumption expenditures compared to West Central region. To show regional disparity in income, **Figure 4** presents an even better picture of how income and expenditure of household is distributed over the. Since violence is related to welfare and might significantly vary from region to region, **Figure 5** can help observe violence across regions.

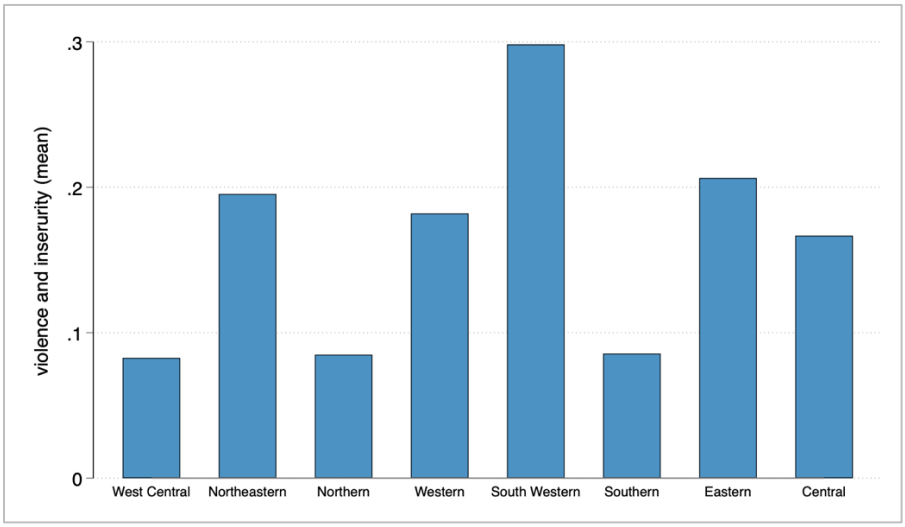
Figure 4.

Distribution of Average Total Income and Average Total Expenditures by Region



Figure 5.

Average Level of Violence by Region



A crucial point that can be observed in these two figures is that in some regions for instance South Western and Central, compared to West Central and Eastern, the average total income is high, and the violence experienced by households is also high, but that does not necessarily mean that higher violence is associated with higher welfare because there are many different factors that positively affect and increase household welfare. In other words, the loss in income in some regions that was caused by violence is much less than the increase in average household income and expenditures that is associated with other factors such as foreign aid, good agricultural land, a small distance from financial centers, and ownership of assets. The case is particularly unique when observing these factors due to the existence of foreign troops in the country. However, this result is consistent with some of the previous studies, for instance Floreani, López-Acevedo, and Rama (2021), that suggested that foreign troops tend to bring more job opportunities and higher foreign aid flows to the areas. This has led to an increase in the level of income and expenditure in the households located in those regions. In other words, the decrease in average total income

due to violence is offset by the increase in average total income due to other economic factors. However, the estimated results from the regression analysis that I carried out show that the overall impact of violence, conditional on other factors, is negative and statistically significant. Overall, the coefficients associated with each region in the model capture the net effect of all these factors and possibly more, reflecting the complex interplay between geographical location and economic prosperity.

Other Findings

The resident code is another variable that presents significant analysis of the violence and income distribution in Afghanistan. The resident code shows whether a household is located in an urban, rural, or are nomadic (Kuchi) who move from one place to another throughout the year. As presented in **Table 4,** the residential code categories 1 (Rural) and 2 (*Kuchi* or Nomadic) in the regression analysis are (-.247) and (-.185) for income, respectively, which are statistically significant. This illustrates that the income levels in rural areas and for nomadic households are significantly lower compared to those in urban households. This impact could be explained by various factors including, but not limited to, the economic opportunity available to households in urban areas and low violence level. In **Figures 6 and 7**, I demonstrate how violence affects the household's welfare in different residential areas. A clear pattern can be observed in **Figure 6** that urban household in Afghanistan has experienced significantly less violence compared to Rural areas and Kuchi, whereas household in urban regions have the highest average income and consumption expenditures among the three regions. Comparable effects can be attained for total expenditures and the coefficient in **Table 5** which shows that this effect is similar on expenditures as well. From this finding, we can conclude that the household in rural residential areas that apparently have experienced more violence have realized less income and consumption and as a

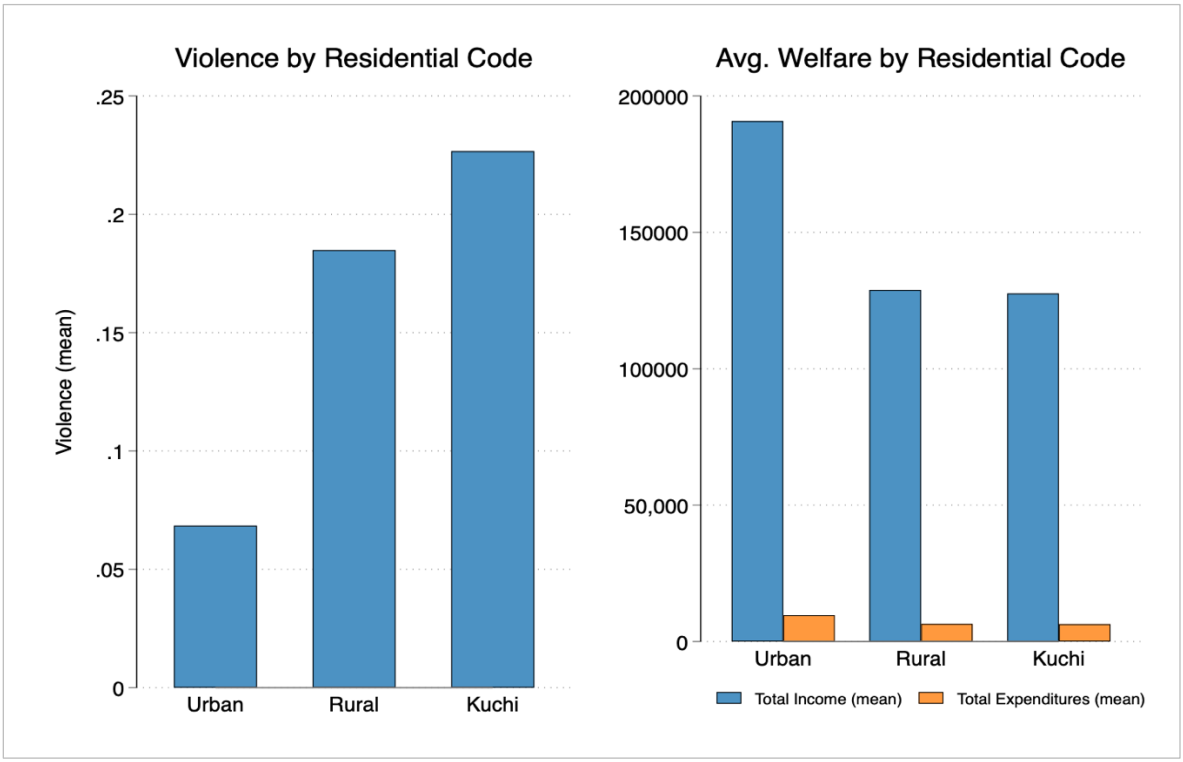
results low welfare, directly supporting our main hypothesis that violence negatively impacts household's welfare.

Figure 6.

Violence Distribution by Resident Areas

Figure 7.

Total Income and Total Expenditures by Resident Areas



In addition, the estimated coefficients for dependency ratio are (-.124) and (-.054) for income and consumption respectively, as shown in **Tables 4** and **Table 5**, which are statistically significant. This negative coefficient measures the effect caused by having one additional dependent member of a household (the non-working members, including children and the elderly) compared to the household with less dependent members. Higher dependency rate not only exposes increased economic pressure over the productive members but causes a reduction in overall wellbeing at the household level by diminishing income which highlights the economic vulnerability of households that is burdened by a higher dependency load. Moreover, the positive

coefficients of (0.128) and (0.088) for income and consumption for the employment status of the household head underline the importance of employment in enhancing household welfare. This coefficient is an indication that the total income for the households with employed household heads is higher compared to those with unemployed heads. Employment could possibly not only improve financial situation but could also gain access to the social networks, health dividends, and other resources that enhance the welfare of the household in general. In other words, this finding further supports those policies that are geared to increase opportunities for employment as a means to economic security and poverty alleviation.

Conclusion

The purpose of this study is to examine the impact of violence and insecurity on Afghan households' welfare. Using over 60,000 observations from three waves of household level data from the Afghanistan Living Condition Survey (ALCS), I measure household welfare using two proxy indicators, the household income and expenditures. The results highlight major and significant influence that violence has on Afghan household welfare. Due to prolonged violence and instability, Afghan households have experienced substantial economic loss and social instability. However, data on conflict is limited as fieldwork is difficult in the context of Afghanistan. I used the households' self-reported measure for violence and insecurity during the survey years.

I employ various techniques including descriptive analysis and OLS regression to explore and assess the causal link between conflict and household welfare variables conditional on several demographic, socioeconomic, and regional factors. The results presented in this study are consistent with the findings of the previous literature revealing that that violence has direct and

indirect negative impacts on different dimensions of household welfare, including infrastructure destruction, displacement, and the abandonment of agricultural fields. The results presented in this study strengthen and broaden our understanding of the impact of violence on household welfare in the unique context of Afghanistan.

The estimated results reveal that there is a significant link between violence and household welfare, with households who have experienced violence or instability earning significantly less total income. To be precise, households that experienced conflict realized about 7.3% less income. Similarly, the results show a reduction in the household consumption expenditures by 2.1% for households that experienced violence. This finding supports the theory that protracted conflict in Afghanistan has negative impacts household welfare either directly or indirectly. This study also provides insights into the geographical distribution of violence and its variations in relation to household welfare outcomes. I find that violence and household welfare highly vary from region to region, but the distribution of violence and household welfare largely overlap, with highly volatile and insecure areas realizing significantly less income and less spending in consumption. Overall, the findings highlight the need for policies and interventions that aim to address the specific issues Afghan households during protracted violence and instability, with an emphasis on providing targeted aid and support for impacted areas and vulnerable groups.

Moreover, the study emphasizes the complex interactions among violence, household demographics, socioeconomic variables, and regional differences. Given the frequent flow of foreign aid, particularly in severely afflicted areas, it acknowledges that the relationship between violence and household welfare in Afghanistan is complex and multidimensional due to other confounding factors. Policymakers, development experts, and academics interested in addressing violence, insecurity, and household welfare in conflict-affected areas can benefit from the

implications of this study, as it explores a unique context of Afghanistan. The results are based on empirical data that can guide the planning and execution of focused interventions aimed to help effectively recover from the damages caused by violence and foster the recovery and prosperity of Afghan households.

References

Arias, M.A., Ibáñez, A.M., Zambrano, A., 2019. Agricultural production amid conflict: Separating the effects of conflict into shocks and uncertainty. World Development 119, 165–184. https://doi.org/10.1016/j.worlddev.2017.11.011

Barrett, Philip. "The fiscal cost of conflict: Evidence from Afghanistan 2005-2016." IMF Working Papers, vol. 18, no. 204, 2018, p. 1, https://doi.org/10.5089/9781484374931.001.

Becker, G. S. (1962). Investment in Human Capital: A Theoretical Analysis. *Journal of Political Economy*, *70*(5, Part 2), 9–49. https://doi.org/10.1086/258724

Blomberg, S.B., Hess, G.D., 2012. The Economic Welfare Cost of Conflict: An Empirical Assessment, in: Garfinkel, M.R., Skaperdas, S. (Eds.), The Oxford Handbook of the Economics of Peace and Conflict. Oxford University Press, p. 0. https://doi.org/10.1093/oxfordhb/9780195392777.013.0018

Bove, V., & Gavrilova, E. (2014). Income and Livelihoods in the War in Afghanistan. *World Development*, *60*, 113–131. https://doi.org/10.1016/j.worlddev.2014.03.021

Central Statistics Organization (CSO), Ministry of Public Health (MoPH), and ICF. 2017. Afghanistan Demographic and Health Survey 2015. Kabul, Afghanistan: Central Statistics Organization.

CSO (2019). *Living conditions survey 2013-2014*. IHSN Survey Catalog. https://catalog.ihsn.org/index.php/catalog/6557

D'Souza, A., Jolliffe, D., 2013. Conflict, food price shocks, and food insecurity: The experience of Afghan households. Food Policy 42, 32–47. https://doi.org/10.1016/j.foodpol.2013.06.007

Floreani, V. A., López-Acevedo, G., & Rama, M. (2021). Conflict and Poverty in Afghanistan's Transition. *The Journal of Development Studies*, *57*(10), 1776–1790. https://doi.org/10.1080/00220388.2021.1945040

Bove, V., & Gavrilova, E. (2014). Income and Livelihoods in the War in Afghanistan. *World Development*, *60*, 113–131. https://doi.org/10.1016/j.worlddev.2014.03.021

D'Souza, A., & Jolliffe, D. (2013). Conflict, food price shocks, and food insecurity: The experience of Afghan households. *Food Policy*, *42*, 32–47. https://doi.org/10.1016/j.foodpol.2013.06.007

Floreani, V. A., López-Acevedo, G., & Rama, M. (2021). Conflict and Poverty in Afghanistan's Transition. *The Journal of Development Studies*, *57*(10), 1776–1790. https://doi.org/10.1080/00220388.2021.1945040

Justino, P. (2011). The Impact of Armed Civil Conflict on Household Welfare and Policy. *IDS Working Papers*, *2011*(384), 1–38. https://doi.org/10.1111/j.2040-0209.2011.00384_2.x

Appendix

Table A1.

Detailed Summary Statistics for individual survey year

	2011/12		2013/14		2016/17	
	mean	sd	mean	sd	mean	sd
HH income (AFN)	135,107	126,388	143,793	137,176	138,159	117,517
HH Expenditures (AFN)	6,186	4,781	7,498	5,181	7,371	5,180
Violence (0/1, 1=yes)	.208	.406	.158	.365	.133	.339
HH Head education (years)	2.855	4.826	2.813	4.803	3.15	4.964
HH head age (years)	41.323	13.472	42.496	13.907	42.79	13.708
Household size (count)	7.641	3.34	7.572	3.346	7.836	3.403
HH head sex (0/1, 1=Male)	.994	.079	.991	.095	.99	.102
HH head employment (0/1, 1=yes)	.834	.372	.778	.416	.759	.428
Dependency ratio	1.293	.893	1.258	.907	1.273	.89
Opium cultivation at the district (*Jeribs*)	42.77	147.87	41.19	140.44	37.74	127.56
Region (0/1, 1=West Central)	.087	.281	.078	.269	.082	.275
Region: 0/1, 1=Northeastern)	.148	.355	.143	.351	.151	.358
Region: 0/1, 1= Northern)	.145	.352	.143	.35	.15	.357
Region: 0/1, 1=Western)	.092	.289	.098	.297	.105	.307
Region: 0/1, 1=South Western)	.144	.351	.137	.344	.12	.325
Region: 0/1, 1=Southern)	.115	.319	.106	.308	.101	.301
Region: 0/1, 1=Eastern)	.12	.325	.11	.313	.096	.295
Region: 0/1, 1=Central)	.149	.356	.184	.387	.194	.395
Residence code (0/1, 1=urban)	.135	.341	.167	.373	.193	.394
Residence code (0/1, 1=rural)	.839	.367	.804	.397	.779	.415
Residence code (0/1, 1=*Kuchi*)	.026	.16	.028	.166	.028	.165
Observations	20,494		20,484		19,501	

Figure 1A.

Household Income & Consumption expenditures during different survey years.

